

Assignment 7

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Q1

Code :

```
f1 = @(x)(sin(x) + cos(x));
x1 = [1];
g1 = gradientF(f1, x1)

f2 = @(x)(x(1)^2 + x(2)^2);
x2 = [1; 2];
g2 = gradientF(f2, x2)

f3 = @(x)(sin(x(2))*(x(1)^2)*cos(x(3)));
x3 = [1; 2; 3];
g3 = gradientF(f3, x3)

f4 = @(x)(x(1) + x(2)*cos(x(3))+ sin(x(4)));
x4 = [1; 2; 3; 4];
g4 = gradientF(f4, x4)

function grad = gradientF(f, point)

    h = 1e-4;           % Fixed perturbation step size
    n = length(point);  % Number of variables
    grad = zeros(1, n); % Initialize gradient vector

    for i = 1:n
        perturbed_point = point;
        perturbed_point(i) = perturbed_point(i) + h;
        f_original = f(point);
        f_perturbed = f(perturbed_point);

        grad(i) = (f_perturbed - f_original) / h;
    end
end
```

Output:

g1 =

-0.3012

$g^2 =$

2.0001 4.0001

$g^3 =$

-1.8005 0.4120 -0.1283

$g^4 =$

1.0000 -0.9900 -0.2821 -0.6536

Q2 b

Code:

```
function newtons_method_rosenbrock()
    x = [0; 0];

    for k = 1:2
        grad = rosenbrock_gradient(x);
        H = rosenbrock_hessian(x);

        x = x - H \ grad;

        fprintf('Iteration %d: x = [%f, %f]\n', k, x(1), x(2));
    end
end

function grad = rosenbrock_gradient(x)
    x1 = x(1);
    x2 = x(2);
    grad = [-400 * x1 * (x2 - x1^2) - 2 * (1 - x1);
            200 * (x2 - x1^2)];
end

function H = rosenbrock_hessian(x)
    x1 = x(1);
    x2 = x(2);
    H = [1200 * x1^2 - 400 * x2 + 2, -400 * x1;
         -400 * x1, 200];
end
```

Output:

Iteration 1: x = [1.000000, 0.000000]

Iteration 2: x = [1.000000, 1.000000]

Q2 c

Code:

```
function gradient_descent_rosenbrock()
    x = [0; 0];
    alpha = 0.05;

    for k = 1:2
        grad = rosenbrock_gradient(x);

        x = x - alpha * grad;

        fprintf('Iteration %d: x = [%f, %f]\n', k, x(1), x(2));
    end
end

function grad = rosenbrock_gradient(x)
    x1 = x(1);
    x2 = x(2);
    grad = [-400 * x1 * (x2 - x1^2) - 2 * (1 - x1);
            200 * (x2 - x1^2)];
end
```

Output:

Iteration 1: x = [0.100000, 0.000000]

Iteration 2: x = [0.170000, 0.100000]

Q3

Code:

```
function conjugate_gradient_rosenbrock()
    x = [-2; 2];
    tol = 1e-6;
    max_iter = 1000;
    k = 0;
    grad = rosenbrock_gradient(x);
    d = -grad;

    while norm(grad) > tol && k < max_iter
        alpha = secant_line_search(@rosenbrock_function, x, d);

        x_new = x + alpha * d;

        grad_new = rosenbrock_gradient(x_new);

        if mod(k, 6) == 0
            d = -grad_new;
        else
            beta_k = (grad_new' * (grad_new - grad)) / (grad' * grad);
            d = -grad_new + beta_k * d;
        end

        x = x_new;
        grad = grad_new;
        k = k + 1;

        fprintf('Iteration %d: x = [%f, %f], f(x) = %f\n', k, x(1), x(2),
rosenbrock_function(x));
    end

    fprintf('Conjugate gradient converged in %d iterations to x = [%f, %f]\n', k,
x(1), x(2));
end

function f = rosenbrock_function(x)
    x1 = x(1);
    x2 = x(2);
    f = 100 * (x2 - x1^2)^2 + (1 - x1)^2;
end

function grad = rosenbrock_gradient(x)
    x1 = x(1);
    x2 = x(2);
    grad = [-400 * x1 * (x2 - x1^2) - 2 * (1 - x1);
            200 * (x2 - x1^2)];
end

function alpha = secant_line_search(func, x, d)
    alpha_prev = 0;
    alpha = 1;
    tol = 1e-5;
    max_iter = 100;

    for i = 1:max_iter
        phi_prev = phi(func, x, d, alpha_prev);
```

```

    phi_curr = phi(func, x, d, alpha);

    if abs(phi_curr) < tol
        break;
    end

    alpha_new = alpha - phi_curr * (alpha - alpha_prev) / (phi_curr -
phi_prev);

    alpha_prev = alpha;
    alpha = alpha_new;
end
end

function val = phi(func, x, d, alpha)
    val = rosenbrock_gradient(x + alpha * d)' * d;
end

```

Output:

```

Iteration 1: x = [-1.458463, 2.134879], f(x) = 6.050068
Iteration 2: x = [-1.432722, 2.031531], f(x) = 5.962922
Iteration 3: x = [-1.242628, 1.485891], f(x) = 5.368495
Iteration 4: x = [-0.955130, 0.848523], f(x) = 4.228940
Iteration 5: x = [-0.944993, 0.895971], f(x) = 3.783872
Iteration 6: x = [-0.744618, 0.513079], f(x) = 3.214893
Iteration 7: x = [-0.600724, 0.301440], f(x) = 2.915497
Iteration 8: x = [-0.563812, 0.326536], f(x) = 2.452994
Iteration 9: x = [-0.390679, 0.119210], f(x) = 2.045672
Iteration 10: x = [-0.277219, 0.027148], f(x) = 1.878321
Iteration 11: x = [0.194653, 0.018167], f(x) = 0.687481
Iteration 12: x = [0.185929, 0.029794], f(x) = 0.664992
Iteration 13: x = [0.339102, 0.091173], f(x) = 0.493512
Iteration 14: x = [0.332249, 0.108276], f(x) = 0.446338
Iteration 15: x = [0.474442, 0.205351], f(x) = 0.315195
Iteration 16: x = [0.614535, 0.354356], f(x) = 0.202859
Iteration 17: x = [0.627553, 0.395757], f(x) = 0.139091
Iteration 18: x = [0.729951, 0.521288], f(x) = 0.086245
Iteration 19: x = [0.815375, 0.649978], f(x) = 0.056163
Iteration 20: x = [0.809247, 0.654045], f(x) = 0.036456
Iteration 21: x = [0.895360, 0.794912], f(x) = 0.015515
Iteration 22: x = [1.000195, 1.000440], f(x) = 0.000000
Iteration 23: x = [1.000307, 1.000623], f(x) = 0.000000
Iteration 24: x = [1.000044, 1.000089], f(x) = 0.000000
Iteration 25: x = [1.000044, 1.000090], f(x) = 0.000000
Iteration 26: x = [1.000045, 1.000090], f(x) = 0.000000
Iteration 27: x = [1.000027, 1.000053], f(x) = 0.000000
Iteration 28: x = [1.000020, 1.000041], f(x) = 0.000000
Iteration 29: x = [1.000018, 1.000039], f(x) = 0.000000

```

Iteration 30: $x = [1.000019, 1.000037]$, $f(x) = 0.000000$
Iteration 31: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 32: $x = [1.000010, 0.999993]$, $f(x) = 0.000000$
Iteration 33: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 34: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 35: $x = [1.000003, 0.999997]$, $f(x) = 0.000000$
Iteration 36: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 37: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 38: $x = [0.999998, 0.999998]$, $f(x) = 0.000000$
Iteration 39: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 40: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 41: $x = [0.999996, 0.999793]$, $f(x) = 0.000004$
Iteration 42: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 43: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 44: $x = [1.000002, 0.999997]$, $f(x) = 0.000000$
Iteration 45: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 46: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 47: $x = [1.000001, 0.999998]$, $f(x) = 0.000000$
Iteration 48: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 49: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 50: $x = [0.999997, 0.999999]$, $f(x) = 0.000000$
Iteration 51: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 52: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 53: $x = [0.999996, 1.000004]$, $f(x) = 0.000000$
Iteration 54: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 55: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 56: $x = [1.000000, 0.999997]$, $f(x) = 0.000000$
Iteration 57: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 58: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 59: $x = [1.000000, 1.000000]$, $f(x) = 0.000000$
Iteration 60: $x = [0.999999, 0.999998]$, $f(x) = 0.000000$
Iteration 61: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 62: $x = [1.000010, 0.999993]$, $f(x) = 0.000000$
Iteration 63: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 64: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 65: $x = [1.000003, 0.999996]$, $f(x) = 0.000000$
Iteration 66: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 67: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 68: $x = [0.999998, 0.999998]$, $f(x) = 0.000000$
Iteration 69: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 70: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 71: $x = [0.999998, 0.999997]$, $f(x) = 0.000000$
Iteration 72: $x = [1.000002, 1.000004]$, $f(x) = 0.000000$
Iteration 73: $x = [1.000007, 1.000021]$, $f(x) = 0.000000$
Iteration 74: $x = [1.000010, 1.000020]$, $f(x) = 0.000000$
Iteration 75: $x = [1.000006, 1.000012]$, $f(x) = 0.000000$

Iteration 76: $x = [0.999939, 0.999964]$, $f(x) = 0.000001$
Iteration 77: $x = [1.000006, 1.000012]$, $f(x) = 0.000000$
Iteration 78: $x = [1.000006, 1.000012]$, $f(x) = 0.000000$
Iteration 79: $x = [0.999995, 0.999990]$, $f(x) = 0.000000$
Iteration 80: $x = [0.999995, 0.999990]$, $f(x) = 0.000000$
Iteration 81: $x = [0.999997, 0.999994]$, $f(x) = 0.000000$
Iteration 82: $x = [0.999998, 0.999995]$, $f(x) = 0.000000$
Iteration 83: $x = [1.000160, 1.000392]$, $f(x) = 0.000001$
Iteration 84: $x = [0.999998, 0.999996]$, $f(x) = 0.000000$
Iteration 85: $x = [0.999998, 0.999996]$, $f(x) = 0.000000$
Iteration 86: $x = [1.000022, 0.999985]$, $f(x) = 0.000000$
Iteration 87: $x = [0.999998, 0.999995]$, $f(x) = 0.000000$
Iteration 88: $x = [0.999998, 0.999995]$, $f(x) = 0.000000$
Iteration 89: $x = [0.999999, 0.999997]$, $f(x) = 0.000000$
Iteration 90: $x = [0.999999, 0.999998]$, $f(x) = 0.000000$
Iteration 91: $x = [0.999999, 0.999997]$, $f(x) = 0.000000$
Iteration 92: $x = [1.000002, 0.999997]$, $f(x) = 0.000000$
Iteration 93: $x = [0.999999, 0.999997]$, $f(x) = 0.000000$
Iteration 94: $x = [0.999999, 0.999997]$, $f(x) = 0.000000$
Iteration 95: $x = [1.000001, 0.999998]$, $f(x) = 0.000000$
Iteration 96: $x = [0.999999, 0.999997]$, $f(x) = 0.000000$
Iteration 97: $x = [0.999999, 0.999997]$, $f(x) = 0.000000$
Iteration 98: $x = [0.999998, 0.999999]$, $f(x) = 0.000000$
Iteration 99: $x = [0.999999, 0.999997]$, $f(x) = 0.000000$
Iteration 100: $x = [0.999999, 0.999997]$, $f(x) = 0.000000$
Iteration 101: $x = [0.999999, 0.999999]$, $f(x) = 0.000000$
Iteration 102: $x = [0.999999, 0.999997]$, $f(x) = 0.000000$
Iteration 103: $x = [0.999999, 0.999997]$, $f(x) = 0.000000$
Iteration 104: $x = [1.000013, 0.999991]$, $f(x) = 0.000000$
Iteration 105: $x = [0.999999, 0.999997]$, $f(x) = 0.000000$
Iteration 106: $x = [0.999999, 0.999997]$, $f(x) = 0.000000$
Iteration 107: $x = [1.000005, 0.999996]$, $f(x) = 0.000000$
Iteration 108: $x = [0.999999, 0.999997]$, $f(x) = 0.000000$
Iteration 109: $x = [0.999999, 0.999997]$, $f(x) = 0.000000$
Iteration 110: $x = [0.999999, 0.999999]$, $f(x) = 0.000000$
Iteration 111: $x = [0.999999, 0.999997]$, $f(x) = 0.000000$
Iteration 112: $x = [0.999999, 0.999997]$, $f(x) = 0.000000$
Iteration 113: $x = [0.999999, 0.999997]$, $f(x) = 0.000000$
Iteration 114: $x = [1.000006, 1.000012]$, $f(x) = 0.000000$
Iteration 115: $x = [1.000002, 1.000003]$, $f(x) = 0.000000$
Iteration 116: $x = [1.000002, 1.000004]$, $f(x) = 0.000000$
Iteration 117: $x = [1.000001, 1.000002]$, $f(x) = 0.000000$
Conjugate gradient converged in 117 iterations to $x = [1.000001, 1.000002]$